

Econ 130: SQ #2

Income Tax

1.a. Regarding optimal income taxation, and the model we used in class, explain why the ratio MU_i/MR_i must rise as t (the income tax rate) rises. Note: MU_i = marginal utility of income for the i^{th} person and MR_i is marginal income tax revenue (we assume government always operates on upward slopping part of the Laffer curve).

1b. Explain why the MU_i/MR_i curve must be lower for high income people than for low income people at any given t .

1c. What is the first order condition regarding t when the objective is to set t so as to maximize aggregate utility subject to the constraint that the government raises a given amount of income tax revenue (R)? Explain by giving an example in which this first order condition is NOT satisfied. What does ultimately imply about the tax rate should set on rich vs poor—use graph to explain.

2a. Initially, all income below I^* is untaxed but for any income above this level it is taxed at a marginal tax rate of t^* . Assume that the worker's optimal bundle is at some income level above I^* . Show this on a graph. Indicate on your graph the amount of tax revenue collected from this worker.

2b. Suppose the marginal tax rate is cut to t^{**} but that the level of untaxed income is reduced so as to leave the worker just as well as before the tax change (hint: “just as well off” means leaves worker with same level of utility). Show this outcome on the same graph. Does tax revenue increase or decrease?

3. Suppose that for pre-tax income of $0-I_1$, the tax rate is t_1 , between I_1 and I_2 the tax rate is t_2 , and for all income above I_2 the tax rate is t_3 with $t_3 > t_2 > t_1$. Assume that the worker's optimal bundle is at H_3 hours of work and at an income level above I_2 . Suppose the government replaces this tax system with one that has a *single* tax rate t^* for all income. The tax rate t^* is selected so that the tax revenue collected would be the same at H_3 hours of work (hint: recall that tax revenue is the vertical gap between the no-tax budget line and the relevant post-tax budget line). Show each situation (pre-tax, old tax system, new tax system) on a single graph. Would the worker be better off under the new tax system? Would the government collect more, the same, or less tax revenue under the new tax system?

4. a. The pre-tax wage is \$1/hour, non-labor income is zero and the worker's time endowment is 24 hours per day. Pre-tax income up to \$4 is not taxed, but all income greater than \$4 is taxed at a rate of 10%. Under these circumstances the worker chooses to work 12 hours per day. On a clearly drawn graph sketch in the pre-tax budget, the after-tax budget line and the relevant indifference curve. What is the amount of tax revenue collected?

4b. Suppose the government changes the tax rate such that any income above \$4 is now taxed at a 20% rate. (so it is still the case that the first \$4 of income is not taxed). Suppose the worker ends up choosing to work a number of hours such that the tax revenue collected is the same as in part (a). What are hours worked? Show this on the *same* graph as used for part (a). To get this outcome, what must we assume about the substitution and income effects?

4c. Is it possible that the increase in the tax rate (without changing the cutoff income) could have resulted in no change in hours of work? *BRIEFLY* explain.

4d. Return to part (a). Suppose the government implements the following new tax system: **ALL** income is taxed at a rate of t^* . The tax rate t^* is selected so as to hold tax revenue constant at 12 hours of work. Find the new tax rate t^* . Does the worker increase or decrease hours of work? Use a graph to justify answer and briefly explain. Does the government collect more or less tax revenue from this worker? Again, justify answer with graph.

Public Goods/Tiebout/Property value maximization

1. Assume five consumers (5 residents or voters in a city) with the following MB_i functions (inverse demand functions): $MB_A = 120 - (1/5)q_A$, $MB_B = 110 - (1/5)q_B$, $MB_C = 100 - (1/5)q_C$,

$MB_D = 90 - (1/5)q_D$, and $MB_E = 80 - (1/5)q_E$.

Marginal cost of public good (C) = 150. Assume the median voter model holds and that the costs of providing the public good are evenly divided so that each voter pays $C/5$.

a. Find the efficient quantity of the public good. Show work and show outcome on a graph.

b. Find the amount each voter prefers of the public good. What is the actual quantity produced? In answering this last question, provide an explanation of why the median voter quantity determines the actual quantity.

c. Return to the above equations. Suppose now that $MB_A = 150 - (1/5)q_A$. How would your answers to parts (a) and (b) change? What is the underlying reason for this new, inefficient outcome? Would the actual quantity of the public good change if A's demand was extremely high relative to other voters, for example, $MB_A = 2000 - (1/5)q_A$.

d. As a general rule, if demands are not symmetric, when would tend to get underprovision of a public good in this model?

2. Tiebout model. Suppose there are two cities Z and V. City Z has five voters, three of type A,

$MB_A = 120 - (1/5)q_A$, and two of type B, $MB_B = 110 - (1/5)q_B$. For City V, five voters, two of type A, and three of type B.

a. Find the efficient q for each city, the desire amounts of q for each type and the actual q in each city. Is either outcome efficient?

b. Find CS for types A and B in each city.

c. Assuming voters can move freely across cities, explain the voting with one's feet phenomenon. Use the CS information in part (b) to justify your answer. Is the final outcome optimal?

3. What does it mean to say property taxes are capitalized into property values. Assume landowners receive annual rents of R and pay annual taxes of T .

4. (There is a chance we won't cover the relevant part for this question). Assume a city of five people with the following preferences for a public good.

$$MB_A = 120 - (1/5)q_A.$$

$$MB_B = 110 - (1/5)q_B.$$

$$MB_C = 100 - (1/5)q_C.$$

$$MB_D = 90 - (1/5)q_D.$$

$$MB_E = 80 - (1/5)q_E.$$

$$C = 150$$

For convenience assume one period. Show that the q at which *aggregate property value* in the city is maximized is the same as the efficient quantity of the public good.

5. Clarke-Groves mechanism. Suppose there are two people in this economy. The marginal benefit schedules for some pure public good are given by $MB_1 = 10 - Q_1$ for individual one and $MB_2 = 16 - Q_2$ for individual two. The marginal cost of providing the good is constant at \$8.

a. Find the optimal quantity for this public good. Show result on a graph.

b. Assume a private, competitive market for this good with a price equal to $MC = \$8$. To illustrate the free-rider problem, suppose each person believes that the other person will buy the good up the quantity where that individual's marginal value (given P in the equation) equals marginal cost. What is the outcome in this case?

c. What is consumer 1's net total gain (U_1) under the Clarke-Groves mechanism assuming both consumers report their MB schedules truthfully? Suppose consumer 1 now "lies" (& consumer 2 reports truthfully) by reporting a MB schedule of $MB_1^* = 8 - Q_1$. Demonstrate that consumer 1 does not have an incentive to do so under a Clarke-Groves Tax.

6. Unanimity Rules and Lindahl pricing. Assume two users of a public good, A & B, with $MB_A = 24 - Q_A$ and $MB_B = 18 - Q_B$. Assume $c = 14$, with S_A being A's *share* of the cost (that is, the tax or price imposed on A expressed as %), with $S_A + S_B = 1$ and $S_A \cdot c + S_B \cdot c = 14$. The quantity produced will be the amount *unanimously* agreed upon by both. If the two consumers of the public good differ on desired quantities of the good, then the government (think of as the auctioneer) will change S_A and S_B until we have unanimous agreement.

a. What is the efficient Q ?

b. Suppose start with 50% share for A. Fill in row one of the table below. If they disagree on quantities, suppose the government changes the share, raising it on A to, say, 0.60. Fill in

row two. Suppose row three is equilibrium outcome, fill in row three (three decimal places for shares).

S_A (in % terms)	S_B (in % terms)	c	$S_A * c$	$S_B * c$	A's desired amount at that price, based on $MB_A = 24 - Q_A$	B's desired amount at the price, based on $MB_B = 18 - Q_B$
0.5	0.5	\$14				
0.6	0.4	\$14				
		\$14				

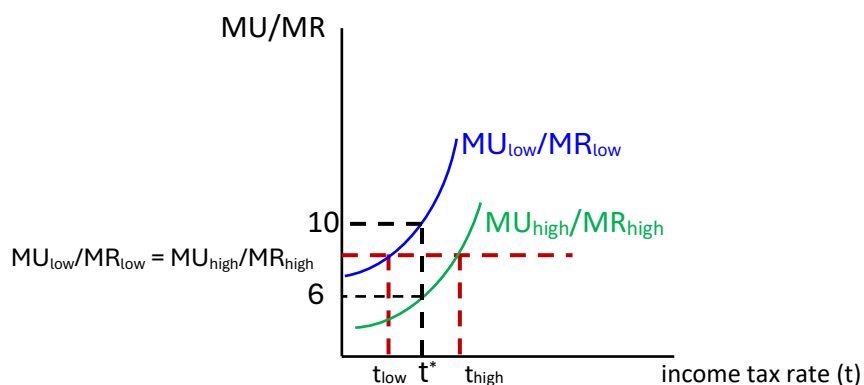
c. Draw a graph showing Lindal prices.

ANSWERS

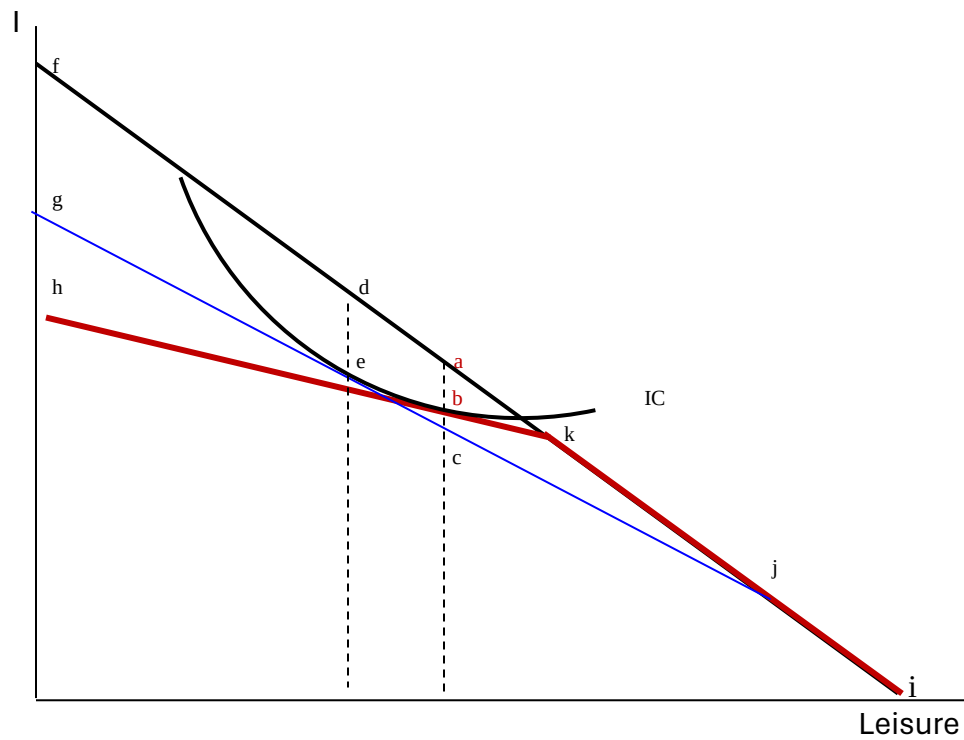
1a. Suppose t rises, person has less after-tax income, thus moves down U function (to where slope is steeper) and therefore MU_i , the slope of the U curve, rises. Also, as t rises we move up the R curve (recall, we assumed we were on the positive sloped side) and MR_i falls. Combining these two outcomes, it follows that MU_i/MR_i must be rising.

1b. A high income person is further up the U function, this person has a lower MU (smaller numerator)—notice we are assuming each person has the same U curve. Also, raising t will collect more revenue from high person than a low income person [example, t is 10%, rich has \$100, poor has \$20 $\rightarrow$ collect \$10 from rich and \$2 from poor; now suppose t is increased to 20%—would get collect \$20 from rich—a \$10 increase in tax revenue—and would now collect \$4 from poor—a \$2 increase in tax revenue from poor. Thus, for the same increase in t , get more additional revenue from rich]. Therefore, the MR (denominator) is higher for a rich person—so it must be if the numerator is lower and the denominator is higher for a high income person then it follows that $(MU_{high}/MR_{high}) < (MU_{low}/MR_{low})$.

1c. The first order condition for optimization problem is $MU_{low}/MR_{low} = MU_{high}/MR_{high}$. Suppose it is not satisfied, for example, suppose that $MU_{low}/MR_{low} = 10 > MU_{high}/MR_{high} = 6$. In this case, suppose the government raised the tax rate on the high income group such that it collected \$1 more tax revenue from them (that is, $MR_{high} = \$1$), and lowered the tax rate on low income group such that it collected \$1 less tax revenue from them (that is, $MR_{low} = \$1$ in absolute value terms). So it follows that total tax revenue is unchanged. But total U is not: the increase in the tax rate on the high income group will cause their utility to fall by 6, and the cut in the tax rate on the low income group raise their utility by 10, thus there is a net increase in society's utility of 4 for the same tax revenue. This can be most clearly by looking at the graph below: Suppose we start at t^* . Notice that at this income tax rate, $MU_{low}/MR_{low} = 10 > MU_{high}/MR_{high} = 6$. If we make the above explained tax rate changes, t up for high income and t down for low income, it can be seen that this moves us to the first order condition of $MU_{low}/MR_{low} = MU_{high}/MR_{high}$.



2.



Pre-tax budget line: **ijkadf**.

Original after-tax budget line: **ijkbh**.

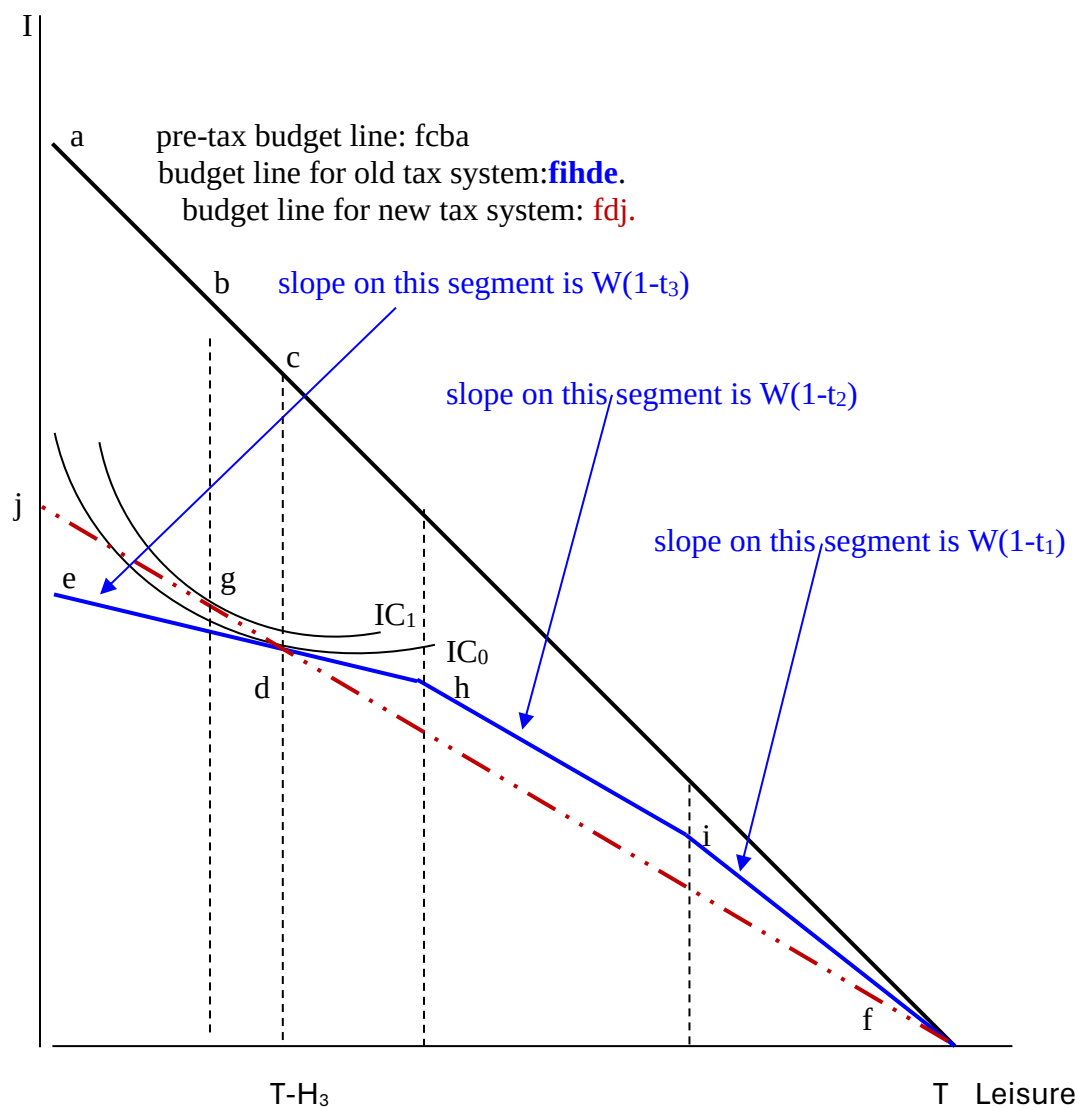
New after-tax budget line: **ijceg**.

Tax revenue under original tax table: **ab**.

Tax revenue under new tax table: **ed**.

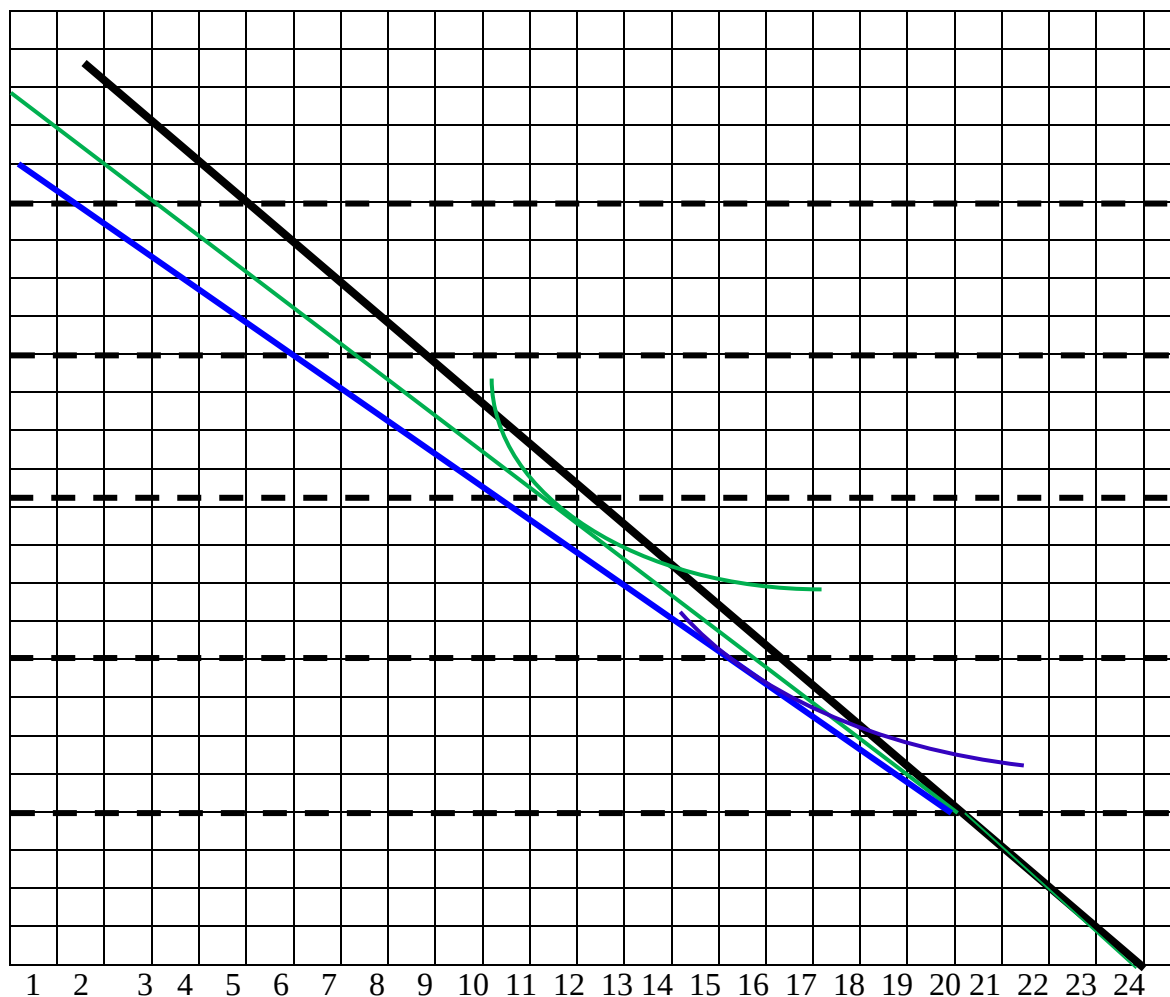
Since it is clear that gap $ca > \text{gap } ab$ and gap $ed > \text{gap } ca$ (the vertical gap gets wider as L falls), then it must be that Gap $ed > \text{gap } ab$. So tax revenue is greater.

3.



As can be seen from the above graph, the worker is better off under the new tax system. Since the new budget line must go through point d (given the restriction that tax revenue be the same at H_3 hours of work), then there will necessarily be bundles that lie above IC_0 and therefore the worker will be better off by selecting one of those bundles. Tax revenue at $T-H_3$ is the same under either tax policy (segment dc), but since the worker ends up at point g under the new tax regime and since $gb > dc$, we can conclude that tax revenue is now higher. Note: $gb > dc$ since the vertical gap widens as L falls.

4a. The pre-tax wage is \$1/hour, non-labor income is zero and the worker's time endowment is 24 hours per day. Pre-tax income up to \$4 is not taxed, but all income greater than \$4 is taxed at a rate of 10%. Under these circumstances the worker chooses to work 12 hours per day. On a clearly drawn graph sketch in the pre-tax budget, the after-tax budget line and the relevant indifference curve. What is the amount of tax revenue collected? answer: vertical axis intercept = \$24 minus tax paid at $H = 24$
tax paid at $H = 24$ is $(t)(w)(H-H^*) = 0.10(1)(24-4) = 0.10(20) = \2 ; at $H = 12$, pre-tax income is \$12 and after tax income is $12 - [(t)(w)(H-H^*)] = 12 - [0.10(1)(12-4)] = 12 - 0.80 = 11.20$; tax is \$0.80



4b. Suppose the government changes the tax rate such that any income above \$4 is now taxed at a 20% rate. (so it is still the case that the first \$4 of income is not taxed). Suppose the worker ends up choosing to work a number of hours such that the tax revenue collected is the same as in part (a). What are hours worked? Show this on the *same* graph as used for part (a). To get this outcome, what must we assume about the substitution and income effects? answer: new vertical axis intercept = \$24 minus tax paid at $H = 24$
tax paid at $H = 24$ is $(t)(w)(H-H^*) = 0.20(1)(24-4) = 0.20(20) = \4 ;

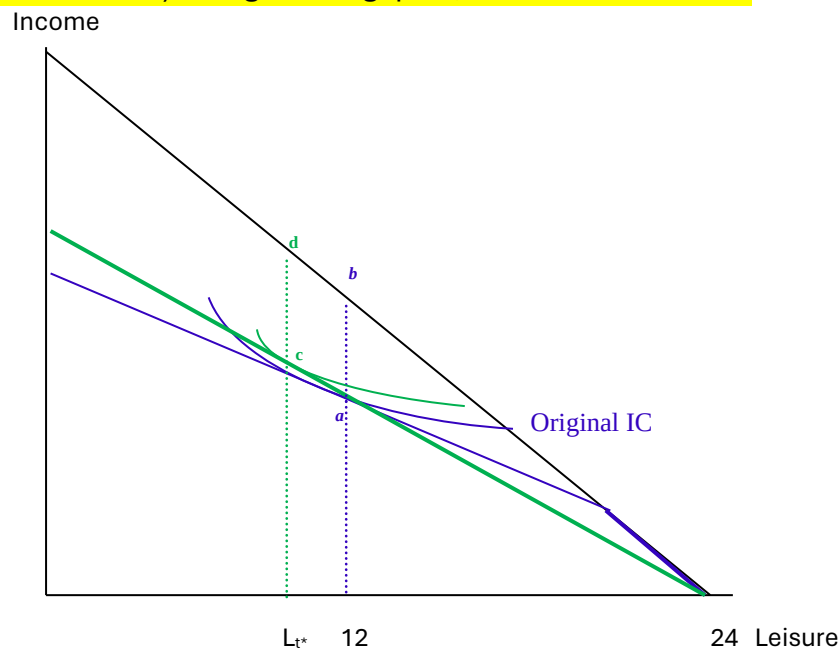
$0.80 = (0.20)(1)(H - 4) = 0.20H - 0.80 \rightarrow H = 1.6/0.2 = 8$ hours (L is now 16). The SE by itself predicts L rises since the after-tax wage is lower (price of leisure is lower). The IE, by itself, predicts L falls since are less wealthy (drop to lower IC). So to get this result that L rises must assume $SE > IE$.

at $H = 12$, pre-tax income is \$12 and after tax income is $12 - [(t)(w)(H-H^*)] = 12 - [0.10(1)(12-4)] = 12 - 0.80 = 11.20$; tax is \$0.80

4c. Is it possible that the increase in the tax rate (without changing the cutoff income) could have resulted in no change in hours of work? *BRIEFLY* explain. Answer: Yes, if $SE = IE$.

4d. Return to part (a). Suppose the government implements the following new tax system: **ALL** income is taxed at a rate of t^* . The tax rate t^* is selected so as to hold tax revenue constant at 12 hours of work. Find the new tax rate t^* . Does the worker increase or decrease hours of work? Use a graph to justify answer and briefly explain. Does the government collect more or less tax revenue from this worker? Again, justify answer with graph. answer: $0.8 = [(t^*)(1)(12)] \rightarrow t^* = 0.80/12 = 0.067$ (6.7%)

At point a (the original tangency), tax revenue is segment ab . The after tax wage (slope of blue BL) is $1(1-.1) = \$0.90$. At the flat tax of 0.067, the after tax wage is 0.93 (approx.). Thus, the price of leisure is higher at point a , the demand for leisure goes down, hours work rises. This can also be seen by the bundles on the green BL that lie above the blue IC. The gap cd (tax revenue under t^*) is larger than gap ab , so tax revenue rises.

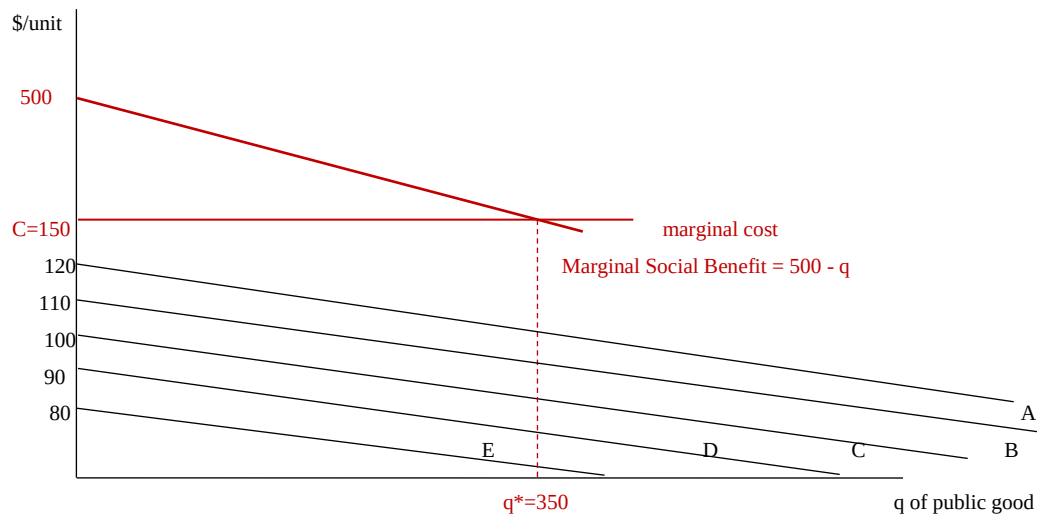


1.a. The efficient level of a public good occurs at the q where $MSB = c$. To get MSB , we need to vertically sum the individual MB schedules:

$$MSB = MB_A + MB_B + MB_C + MB_D + MB_E$$

$$MSB = [120 - (1/5)q] + [110 - (1/5)q] + [100 - (1/5)q] + [90 - (1/5)q] + [80 - (1/5)q] = 500 - q.$$

$$\rightarrow 500 - q = 150 \text{ or } q^* = 350.$$



1b. Next step is to determine the *actual* level of production. Two things we need to do. Find each voter's desired level of q given that each voter pays $c/5$. Then identify the median voter's quantity. Solving for desired levels:

$$MB_A = 120 - (1/5)q_A = c/5 = 150/5 = 30 \rightarrow q_A = 450.$$

$$MB_B = 110 - (1/5)q_B = c/5 = 150/5 = 30 \rightarrow q_B = 400.$$

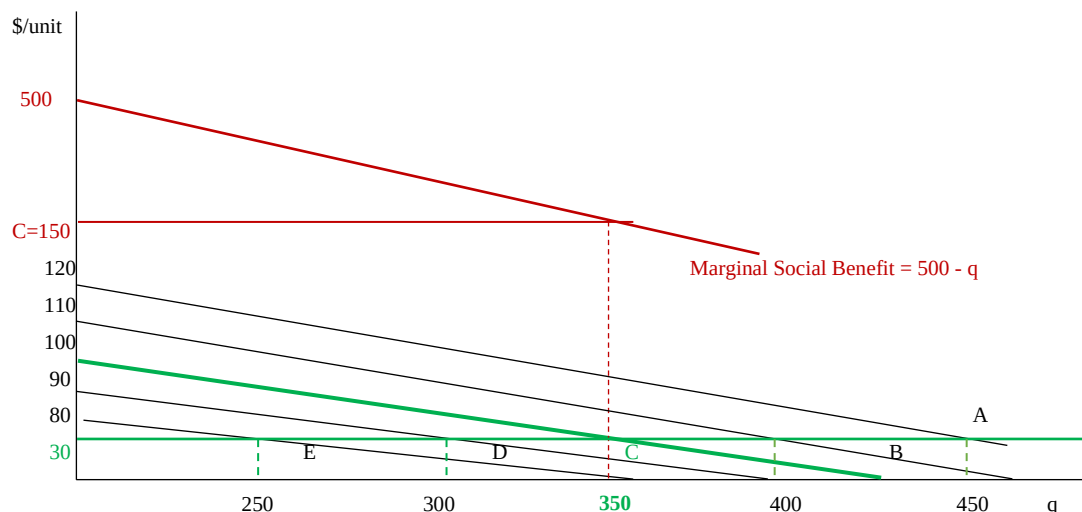
$$\mathbf{MB_C = 100 - (1/5)q_C = c/5 = 150/5 = 30 \rightarrow q_C = 350 \text{ (median voter)}}$$

$$MB_D = 90 - (1/5)q_D = c/5 = 150/5 = 30 \rightarrow q_D = 300.$$

$$MB_E = 80 - (1/5)q_E = c/5 = 150/5 = 30 \rightarrow q_E = 250.$$

The actual amount produced equals the median voter quantity. In this case, the q actually produced equals the efficient quantity.

Logic of the median voter model: consider two candidates who compete by offers of q . Suppose initially candidate #1 offers to produce 380 units if elected and candidate #2 offers to produce 310 units if elected. Candidate #1 will get the votes of A and B since the offer of 380 is closer to their preferred amounts and candidate #2 will get the votes of D and E since an offer of 310 is closer to their preferences. What about voter C? C's preferred amount is 350, candidate #1 offers 380 or 30 units away from C's preferred amount, whereas candidate #2 offers 310 or 40 units from C's preferred amount. Since #1 is closer, C would vote for #1 (this is equivalent to saying the median voter votes for the candidate that maximizes their consumer surplus). However, #2 can counter by offering 322 units if elected—would still get D and E's votes, and since this offer is now 28 units from C's preferred amount #2 would now get C's vote. This round-by-round competition, according to median voter model, would cause q to move closer and closer to 350. End result is the actual q produced is equal to the median voter quantity.



Next step is to determine the *actual* level of production. Two things we need to do. Find each voter's desired level of q given that each voter pays $c/5$. Then identify the median voter's quantity.

Solving for desired levels:

$$MB_A = 120 - (1/5)q_A = c/5 = 150/5 = 30 \rightarrow q_A = 450.$$

$$MB_B = 110 - (1/5)q_B = c/5 = 150/5 = 30 \rightarrow q_B = 400.$$

$$\mathbf{MB_C = 100 - (1/5)q_C = c/5 = 150/5 = 30 \rightarrow q_C = 350 \text{ (median voter)}}$$

$$MB_D = 90 - (1/5)q_D = c/5 = 150/5 = 30 \rightarrow q_D = 300.$$

$$MB_E = 80 - (1/5)q_E = c/5 = 150/5 = 30 \rightarrow q_E = 250.$$

$$1.c. \text{ MSB} = MB_A + MB_B + MB_C + MB_D + MB_E$$

$$\text{MSB} = [150 - (1/5)q] + [110 - (1/5)q] + [100 - (1/5)q] + [90 - (1/5)q] + [80 - (1/5)q] = 530 - q.$$

$$\rightarrow 530 - q = 150 \text{ or } q^* = 380. \text{ The efficient quantity would now be higher at 380 units.}$$

$$MB_A = 150 - (1/5)q_A = c/5 = 150/5 = 30 \rightarrow q_A = 600.$$

$$MB_B = 110 - (1/5)q_B = c/5 = 150/5 = 30 \rightarrow q_B = 400.$$

$$\mathbf{MB_C = 100 - (1/5)q_C = c/5 = 150/5 = 30 \rightarrow q_C = 350 \text{ (median voter)}}$$

$$MB_D = 90 - (1/5)q_D = c/5 = 150/5 = 30 \rightarrow q_D = 300.$$

$$MB_E = 80 - (1/5)q_E = c/5 = 150/5 = 30 \rightarrow q_E = 250.$$

A's desire quantity would change, but there is no change in median voter q , hence the actual quantity would still be at 350 (same logic as given in the answer to 7b). In this case, we get underprovision of the public good.

The voting process does not take into account the intensity of preferences of non-median voters. In this case, A's demand increased for this good, but A's vote still shows up 1 or 0 ("yes" or "no"). So even if $MB_A = 2000 - (1/5)q_A$, C would still be the median voter and therefore still get 350 units produced.

1.d. We would get underprovision when the mean demand $>$ median demand. Take the above example from 7c.

$$\text{Mean demand in part (c) is } (600 + 400 + 350 + 300 + 250) / 5 = 380 > \text{median demand} = 350.$$

2a. efficient quantities:

City Z: $MSB = [120 - (1/5)q] + [120 - (1/5)q] + [120 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] = 580 - q$.
 $580 - q = 150 \rightarrow q^* = 430$.

City V: $MSB = [120 - (1/5)q] + [120 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] = 570 - q$.
 $570 - q = 150 \rightarrow q^* = 420$.

Desired amounts:

$MB_A = [120 - (1/5)q_A] = 150/5 = 30 \rightarrow q_A = 450$

$MB_B = [110 - (1/5)q_B] = 150/5 = 30 \rightarrow q_B = 400$

Median voter:

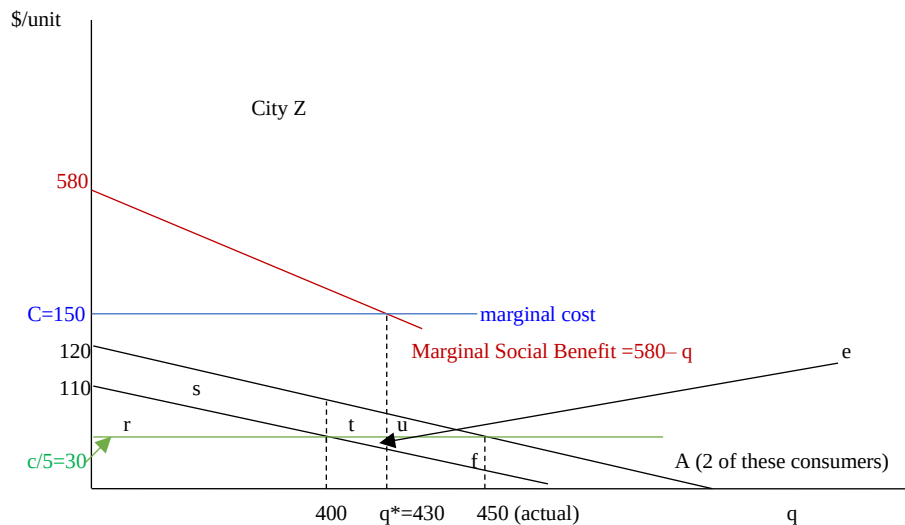
City Z, it is voter A, so actual q is 450.

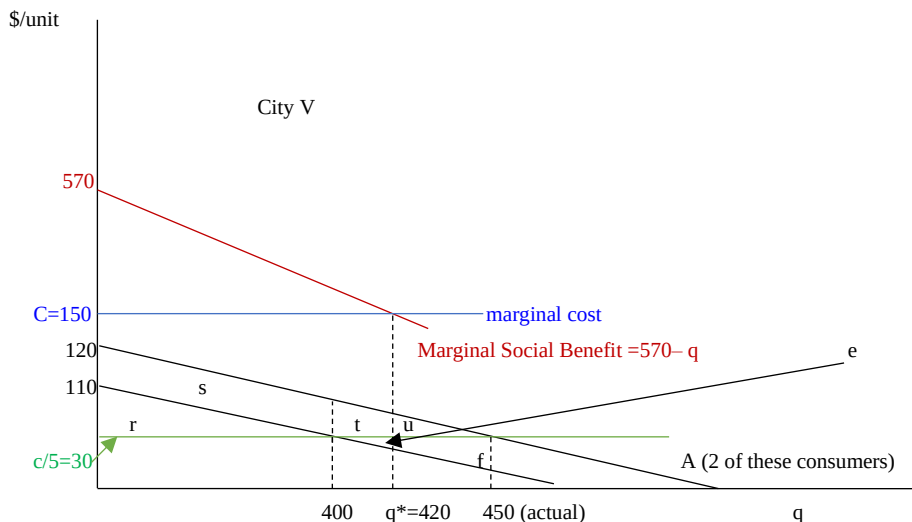
City V, it is voter B, so actual q is 400.

Neither outcome is efficient. In city Z: efficient $q = 430 < \text{actual } q = 450$. In city V: efficient $q = 420 > \text{actual } q = 400$.

2b. City Z: CS of A is $rstu$. CS of B is $r-ef$.

City V: CS of A is rs . CS of B is r .





2c. For type A, $CS \text{ in } Z = rstu > CS \text{ in } V = rs$. Therefore, the two A voters in V have an incentive to move to Z. For type B, $CS \text{ in } Z = r - ef < CS \text{ in } V = r$. Therefore, the two B voters in Z have an incentive to move to V.

As a result, all five of type A end up in city Z and all five of type B end up in city V.

City Z: $MSB = [120 - (1/5)q] + [120 - (1/5)q] + [120 - (1/5)q] + [120 - (1/5)q] + [120 - (1/5)q] = 600 - q$.
 $600 - q = 150 \rightarrow q^* = 450$.

City V: $MSB = [110 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] + [110 - (1/5)q] = 550 - q$.

$550 - q = 150 \rightarrow q^* = 400$.

Desired amounts:

$MB_A = [120 - (1/5)q_A] = 150/5 = 30 \rightarrow q_A = 450$

$MB_B = [110 - (1/5)q_B] = 150/5 = 30 \rightarrow q_B = 400$

Median voter:

City Z, it is voter A, so actual q is 450.

City V, it is voter B, so actual q is 400.

Both outcomes are efficient. In city Z: efficient q = 450 = actual q = 450. In city V: efficient q = 400 = actual q = 400.

3. It simply means that property taxes influence the price of the property—or, another way of saying it, property taxes affect the PV of the land. The formal way of expressing it is $V_0 = (R - T)/i$ with V_0 being the present value or current price of the property, R is annual rental the landowner receives (inflow of cash), T is annual property taxes paid by the landowner (outflow of cash), and i is the interest rate. Notice that as T goes up, other things constant, V_0 falls—that is, property taxes are capitalized into the price of land. Also notice that a higher R , other things constant, causes V_0 to rise. Something that would cause R to rise would be additional units of a public good (say, even more clean air, even less crime, more parks, etc.)

4. First, find efficient quantity.

$$MSB = MB_A + MB_B + MB_C + MB_D + MB_E$$

$$MSB = [120 - (1/5)q] + [110 - (1/5)q] + [100 - (1/5)q] + [90 - (1/5)q] + [80 - (1/5)q] = 500 - q.$$

$$\rightarrow 500 - q = 150 \text{ or } q^* = 350.$$

Second, we need to find the rent functions for each person then the aggregate rent function.

$$R \text{ to } A = TB_A = \int MB_A dq = 120q_A - 0.10q_A^2.$$

$$R \text{ to } B = TB_B = \int MB_B dq = 110q_B - 0.10q_B^2.$$

$$R \text{ to } C = TB_C = \int MB_C dq = 100q_C - 0.10q_C^2.$$

$$R \text{ to } D = TB_D = \int MB_D dq = 90q_D - 0.10q_D^2.$$

$$R \text{ to } E = TB_E = \int MB_E dq = 80q_E - 0.10q_E^2.$$

$$\text{Aggregate rents} = [120q - 0.10q^2] + [110q - 0.10q^2] + [100q - 0.10q^2] + [90q - 0.10q^2] + [80q - 0.10q^2]$$

$$\text{Aggregate rents} = 500q - 0.50q^2.$$

Third, Aggregate property taxes = total cost of producing q units of public good = $150q$.

Fourth, Aggregate property values = Aggregate rents – aggregate property taxes

$$\text{Aggregate property values (V)} = [500q - 0.50q^2] - [150q] = 350q - 0.50q^2.$$

The q that maximizes V is given by

$$dV/dq = 350 - q = 0 \text{ (F.O.C.)}, \text{ or } q = 350, \text{ same as efficient } q.$$

5.a. We need to *vertically* sum each curve. Therefore, we must first solve for the variable on the vertical axis (that is, MB). $MB_1 = 10 - Q$ and $MB_2 = 16 - Q$ or $MB_{\text{social}} = 26 - 2Q$. (for $Q < 10$). Setting MB_{social} (or summed MBs) equal to MC and solving for Q gives us the socially optimal quantity. $26 - 2Q = 8$ or $Q = 9$.

5b. Consumer one believes that consumer two will buy the good up to 8 units (since $16 - Q = 8$ or 8 units). Consumer one's optimal amount is given by $10 - Q = 8$ or 2. With anticipation of consumer two buying 8 units, consumer one will not purchase any units.

(Alternatively, we can say that person one will not buy any more units since the marginal benefit of a 9th unit ($= 10 - Q = 10 - 9 = \$1$) is less than the price of \$8.)

Consumer two believes that consumer one will buy up to 2 units (since $10 - Q = 8$ or $Q = 2$). So person two will purchase 6 units (up to the point where marginal benefit equals \$8 $\leftarrow 16 - Q = 8$)

The outcome is that one does not buy any and two buys 6 units, so 6 units are produced and consumed. This is below the optimal quantity of 9 units. The reason, of course, is that each individual is free-riding on the other's purchases. This happens because once the good is produced, a consumer can use it whether he or she paid for it or not.

5c. Answer: First, find quantity government would produce given truthful reporting. We have already done that in part (a), so $Q = 9$.

Second, **figure out the Clarke-Groves tax** →

The Clarke-Groves Tax is given by $T_1 = TC - TB_2 = (\$8)(9) - [16Q - 0.5Q^2] = \$72 - [16(9) - 0.5(9)^2] = 72 + 40.5 - 144 = -31.5$ (a subsidy of 31.5). (MATH NOTE: for $MB_2 = 16 - Q$, the Total benefit schedule is $TB_2 = \int 16 - Q dQ = 16Q - 0.5Q^2$).

Third, **figure out net total benefits** →

$$U_1 = TB_1 - T_1 = 10Q - 0.5Q^2 - (-31.5) = 10(9) - 0.5(9)^2 + 31.5 = 121.5 - 40.5 = \$81.$$

Answer: First, find the quantity produced by the government given these assumptions.

Vertically adding the MB schedules gives us $MB_{\text{social}}^* = (8-Q) + (16-Q) = 24 - 2Q$. Setting this equal to MC and solving for Q gives us $24 - 2Q = 8$ or **Q = 8**.

Second, **figure out the Clarke-Groves tax at this Q** →

$T_1 = TC - TB_2 = (\$8)(8) - [16Q - 0.5Q^2] = \$64 - [16(8) - 0.5(8)^2] = 64 + 32 - 128 = -32$ (a subsidy of 32).

Third, figure out net total benefits →

$U_1 = TB_1 - T_1 = 10Q - 0.5Q^2 - (-32) = 10(8) - 0.5(8)^2 + 32 = 112 - 32 = \80 . This is lower than the net total benefits 1 receives when reporting truthfully.

6a. Marginal Social Benefit (MSB) = $MB_A + MB_B = (24-Q) + (18-Q) = 42 - 2Q$ (for $Q < 18$).

Setting $MSB = c$ and solving for Q: $42 - 2Q = 14 \rightarrow Q^* = 14$.

6b.

S_A (in % terms)	S_B (in % terms)	c	$S_A * c$	$S_B * c$	A's desired amount at that price, based on $MB_A = 24 - Q_A$	B's desired amount at the price, based on $MB_B = 18 - Q_B$
0.5	0.5	\$14	\$7	\$7	17	11
0.6	0.4	\$14	\$8.4	\$5.6	15.6	12.4
.714	.286	\$14	\$10 (rounding)	\$4 (rounding)	14	14

C.

